

Abhiram Kollipara

(he/him) | abhiramkollipara05@gmail.com | 508-310-3375

 akollipara.com —  linkedin.com/in/abhiram-kollipara —  github.com/AbhiK1212

EDUCATION

University of Massachusetts Amherst

B.S in Computer Science

Amherst, MA

Expected Dec 2026

- **Courses:** Algorithms, Data Structures, Programming Methodology, Web Programming, ML, AI, Systems, Databases, IOT

EXPERIENCE

Tesla

Software Engineer Intern

Fremont, CA

Jan - May 2026

- Developed **full-stack** Next.js features for internal Sales & Management Platform and AccountOS CRM serving **133K+** users
- Built **Go** and **Java** backend microservices behind the platform's Express/GraphQL BFF, improving response times **~30%**
- Engineered a Python/FastAPI AI service with LangChain/LangGraph and SSE, cutting p95 latency **40%** via Redis caching
- Contributed to distributed system architecture leveraging **Kubernetes** and **Kafka** for scalable event-driven communication

Intuit Credit Karma

Mobile Software Engineer Intern

Charlotte, NC

May - Aug 2025

- Engineered swipeable card stack in **Kotlin/XML** for the Android app main dashboard serving **50M+** monthly active users
- Delivered **+0.13%** lift in Retained Monthly Members (RMM) and achieved **20% CTR**— over 10x the industry benchmark
- Architected **GraphQL** to **MVVM** data pipeline to process **500K+** daily API requests, achieving **<200ms** response times
- Launched iOS-exclusive feature on Android across **25+** portal configurations, generating an estimated **\$2M+** ARR impact
- Revitalized Discover Video feature from **-8%** revenue to neutral via **Darwin A/B** testing, raising user engagement by **22%**

Haver AI

Software Engineer Intern

Harvard Innovation Labs, Boston MA

June - Sept 2024

- Engineered open-source healthcare predictive analysis platform using **React**, **AWS**, **SageMaker** + **Scikit-learn** ML models
- Architected **10+** RESTful APIs with **API Gateway** + **Lambda** (Node.js/Python), integrating with **S3** and **DynamoDB**
- Optimized S3-based data archiving solution with tiered **Glacier** lifecycle policy, reducing long-term storage costs by **~30%**
- Developed **React** components for secure, large-file uploads (up to 5GB) to **S3** using presigned URLs & preview capabilities
- Automated empty S3 folder cleanup using scheduled Lambda functions via **EventBridge**, streamlining storage management

Build UMass

Software Engineer, Aarti Home Team

University of Massachusetts Amherst

Jan 2023 - Present

- Shipped multi-lingual **React Native** app with team of 10 engineers for an Indian NGO supporting 6,400+ abandoned girls
- Improved data load times by **90%** with **MMKV** encrypted storage over AsyncStorage, enhancing UX on weak connections
- Architected **Flask** REST APIs with DynamoDB to serve content, quizzes, and progress for **2,000+** students in pilot testing
- Built **RAG**-powered AI chatbot in Python with **ChromaDB**, delivering context-aware answers from educational resources

PROJECTS

FoundU (Github + Video) | *React Native, Expo, Google Vision API, Node, Express, Multer, Puppeteer, Nodemailer, Cookie-Editor*

- Placed **Top 8** at HackUMass XI, winning the **Wolfram Award (\$2,000)** for a cross-platform mobile app that automates the return of lost student ID cards for **30,000+** students using React Native, Google Vision API, and Nodemailer notifications
- Integrated **Puppeteer** web scraper using cookie-auth to interface with UMass systems, automating student email retrieval

FinSight (Github + Video) | *Python, PyTorch, FastAPI, Docker, MLOps (Prometheus, Grafana, MLflow)*

- Fine-tuned **DistilBERT** model on 2,300+ financial headlines, achieving **macro-F1 of 0.9** on 3-class sentiment classification
- Deployed FastAPI microservice for **700+** req/s workloads with **~60ms** latency and 40% faster responses via **Redis** caching
- Built **MLOps** pipeline with **Prometheus/Grafana** for real-time monitoring of KPIs and **MLflow** for experiment tracking

Spark (Github + Video) | *TypeScript, Next, Express, Docker, FFmpeg, Tailwind, Firestore, Firebase, Pub/Sub, Cloud Run, CI/CD*

- Engineered scalable video-streaming platform on **GCP** with event-driven microservices, handling **1000+** concurrent uploads
- Automated **FFmpeg** transcoding pipeline on Cloud Run, triggered by Pub/Sub for elastic scaling and metadata storage
- Built Next.js web client with **Firebase Auth** and signed URLs to Cloud Storage for real-time status tracking and analytics
- Integrated **CI/CD** with GitHub Actions for automated Cloud Run deployments, reducing deployment time from 15 to **3 min**

SKILLS

Languages: JavaScript, Python, Go, Java, Kotlin, Swift, SQL, C, C++, Dart, HTML/CSS

Frameworks/Libraries: React, Node.js, Next.js, Express, GraphQL, FastAPI, Flask, Spring, PyTorch, Pandas, LangChain

Cloud/Database: Kubernetes, Kafka, AWS, Google Cloud Platform, MySQL/PostgreSQL, Redis, MongoDB

Tools: Git, Docker, Linux, CI/CD, Prometheus, Grafana, MLflow, OpenAI, Jenkins, Jira, Splunk, JUnit, Jest